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VMC Vidyamandir Classes SINCE 1986
IIT JEE | MEDICAL | FOUNDATION

CODE A

Mock JEE Main-2 (CBT) | JEE-2024

Date: 08/01/2024 **Maximum Marks: 300**

Timing: 3:30 PM to 6:30 PM **Duration : 3.0 Hours**

General Instructions

1. The test is of **3 hours** duration and the maximum marks is **300.**
2. The question paper consists of **3 Parts** (Part I: **Physics**, Part II: **Chemistry**, Part III: **Mathematics**). Each Part has **two** sections (Section 1 & Section 2).
3. **Section 1** contains **20 Multiple Choice Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.
4. **Section 2 contains 10 Numerical Value Type Questions Out of which ONLY 5 (any) questions have to be attempted. You will NOT be allowed to attempt the sixth question. If you wish to attempt any other question apart from the five already attempted, then you will have to delete any one response from the five previously answered and then proceed to answer the new one.**
   The answer to each question should be **rounded off to the nearest integer**.
5. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. On completion of the test, the candidate must hand over the Answer Sheet to the **Invigilator** on duty in the Room/Hall. **However, the candidates are allowed to take away this Test Booklet with them**.

Marking Scheme

1. **Section – 1:** +4 for correct answer, –1 (negative marking) for incorrect answer, 0 for all other cases.
2. **Section – 2:** +4 for correct answer, –1 (negative marking) for incorrect answer, 0 for all other cases.

Name of the Candidate (In CAPITALS) : _____

Roll Number : _____

OMR Bar Code Number : _____

Candidate's Signature : _____ Invigilator's Signature _____

PART I : PHYSICS	MARKS: 100

SECTION-1

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. A point charge q is placed at a distance of r from the centre of an uncharged conducting sphere of radius $R(<r)$. The potential at any point on the sphere is:

(A) zero **(B)** $\frac{1}{4\pi\varepsilon_0}.\frac{q}{r}$ **(C)** $\frac{1}{4\pi\varepsilon_0}.\frac{qR}{r^2}$ **(D)** $\frac{1}{4\pi\varepsilon_0}.\frac{q}{R}$

2. Vertical displacement of a plank with a body of mass 'm' on it is varying according to law $y=\sin\omega t+\sqrt{3}\cos\omega t$. The minimum value of ω for which the mass just breaks off the plank and the moment it occurs first after $t = 0$ are given by: (y is positive vertically upwards)

(A) $\sqrt{\frac{g}{2}}, \frac{\sqrt{2}}{6}\frac{\pi}{\sqrt{g}}$ **(B)** $\frac{g}{\sqrt{2}}, \frac{2}{3}\sqrt{\frac{\pi}{g}}$ **(C)** $\sqrt{\frac{g}{2}}, \frac{\pi}{2}\sqrt{\frac{2}{g}}$ **(D)** $\sqrt{2g}, \sqrt{\frac{2\pi}{3g}}$

3. Match the following:

Table-1 **Table-2**

I	Electrical resistance	**(P)**	$\left[M^{-1}L^{-2}T^{4}A^{2}\right]$
II	Capacitance	**(Q)**	$\left[ML^{2}T^{-2}A^{-2}\right]$
III	Magnetic field	**(R)**	$\left[ML^{2}T^{-3}A^{-2}\right]$
IV	Inductance	**(S)**	$\left[MT^{-2}A^{-1}\right]$

(A) $I\to P, II\to Q, III\to S, IV\to R$ **(B)** $I\to R, II\to P, III\to S, IV\to Q$

(C) $I\to P, II\to R, III\to S, IV\to Q$ **(D)** $I\to R, II\to Q, III\to P, IV\to Q$

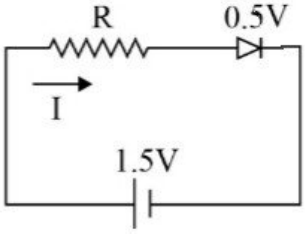
4. The diode used in the circuit shown in figure has a constant voltage drop of 0.5 V at all currents and a maximum power rating of 100 milliwatt. What should be the value of the resistor R, connected in series with the diode, for obtaining maximum current?

(A) 1.5Ω

(B) 5Ω

(C) 6.67Ω

(D) 200Ω



SPACE FOR ROUGH WORK

5. The equation of trajectory of a projectile is $y=\sqrt{3}x-\frac{g x^2}{2}$. The velocity of projection is: (all units are SI)

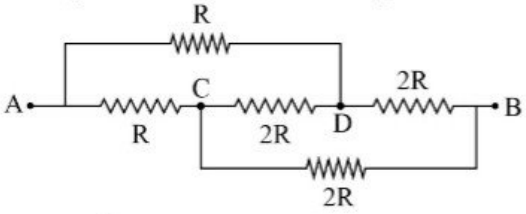
(A) 4 m/s **(B)** 2 m/s **(C)** $\frac{2}{\sqrt{3}}$ m/s **(D)** 20 m/s

6. An excited hydrogen atom returns to the ground state. The wavelength of emitted photon is λ. The principal quantum number of the excited state will be:

(A) $\left(\frac{\lambda R}{\lambda R-1}\right)^{1/2}$ **(B)** $\left(\frac{\lambda R+1}{\lambda R}\right)^{1/2}$

(C) $\left[\lambda R(\lambda R+1)\right]^{1/2}$ **(D)** $\left[\frac{1}{\lambda R(\lambda R+1)}\right]^{1/2}$

7. The equivalent resistance between points A and B in the circuit shown in figure is:



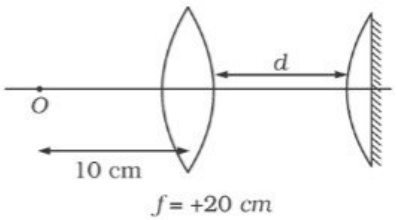
(A) $\frac{R}{2}$ **(B)** $2R$ **(C)** $\frac{3R}{2}$ **(D)** $\frac{6R}{5}$

8. In a hydraulic lift at a service station, the radii of the large and small pistons are in the ratio of 20 : 1. What weight placed on the small piston will be sufficient to lift a car of mass 1200 kg?

(A) 3 kgf **(B)** 30 kgf **(C)** 300 kgf **(D)** 3000 kgf

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9. A convex lens of focal length 20 cm and another plano-convex lens of focal length 40 *cm* are placed co-axially. The plano-convex lens is silvered on plane surface. What should be the distance d (in cm) so that final image of the object 'O' is formed on O itself?



(A) 10 **(B)** 15
(C) 20 **(D)** 25

10. The average kinetic energy of translatory motion of hydrogen molecules at 27°C is $6.2\times10^{-21}J$. The average kinetic energy at 127°C will be:

(A) $30\times10^{-21}J$ **(B)** $8.27\times10^{-21}J$ **(C)** $1.2\times10^{-21}J$ **(D)** $3.1\times10^{-21}J$

11. If R_m is the radius of Moon orbiting round the Earth, a_m the acceleration of Moon towards the centre of Earth, and R_e the radius of Earth. Then a_m is equal to (if g is acceleration due to gravity on the surface of Earth)

(A) $\left(\frac{R_e}{R_m}\right)g$ **(B)** $\left(\frac{R_m}{R_e}\right)g$ **(C)** $\left(\frac{R_m}{R_e}\right)^2 g$ **(D)** $\left(\frac{R_e}{R_m}\right)^2 g$

12. If one mole of a monoatomic gas $\left(\gamma=\frac{5}{3}\right)$ is mixed with one mole of a triatomic gas $\left(\gamma=\frac{4}{3}\right)$, the value of γ for the mixture is:

(A) 1.40 **(B)** 1.44 **(C)** 1.53 **(D)** 3.07

13. Given below are two statements:
Statement I: A body in translatory motion is momentarily at rest when it reverses the direction.
Statement II: A body can not be accelerated if its velocity is zero at a given instant of time.

(A) Both statement I and statement II are true
(B) Both statement I and statement II are false
(C) Statement I is true but statement II is false
(D) Statement I is false but statement II is true

SPACE FOR ROUGH WORK

14. The X-ray beam coming from an X-ray tube will be:

(A) Monochromatic

(B) having all wavelengths smaller than a certain maximum wavelength

(C) having all wavelengths larger than a certain minimum wavelength

(D) having all wavelengths lying between a minimum and a maximum wavelength

15. Green light of wavelength 5100Å from a narrow slit is incident on a double slit. If the overall separation of 10 fringes on a screen 3m away is 3 cm, then the double slit separation is:

(A) 0.051 mm **(B)** 0.51 mm **(C)** 5.1 mm **(D)** 5.1 cm

16. **Assertion:** An object moving with a velocity of magnitude 10 m/s is subjected to a uniform acceleration $2m/s^2$ at right angle to the initial motion. Its velocity after 5s has a magnitude nearly 14 m/s.

Reason: The equation $\vec{v} = \vec{u} + \vec{a}t$ can be applied to obtain $\vec{v}$ if $\vec{a}$ is constant.

(A) Both Assertion and Reason are correct and Reason is correct explanation of Assertion

(B) Both Assertion and Reason are correct but Reason is not correct explanation of Assertion

(C) Assertion is correct but Reason is incorrect

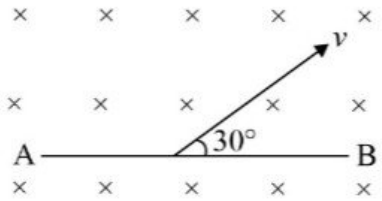
(D) Assertion is incorrect and Reason is correct

17. In a photoelectric cell, the wavelength of incident light is changed from $4000\text{Å}\ to\ 3600\text{Å}$. The change in stopping potential will be:

(A) 1.14 V **(B)** 0.24 V **(C)** 0.35 V **(D)** 0.44 V

SPACE FOR ROUGH WORK

18. A conducting rod AB of length $l = 1\,m$ is moving at a velocity $v = 4\,m/s$ making an angle 30° with its length. A uniform magnetic field $B = 2T$ exists in a direction perpendicular to the plane of motion. Then:



(A) $V_A - V_B = 8V$ **(B)** $V_A - V_B = 4V$ **(C)** $V_B - V_A = 8V$ **(D)** $V_B - V_A = 4V$

19. The rear side of a truck is open and a box of mass 20 kg is placed on the truck 4m away from the open end. If $\mu = 0.15$ and $g = 10\,ms^{-2}$ and the truck starts from rest with an acceleration of $2ms^{-2}$ on a straight road, then the box will fall off the truck when the truck is at a distance of x metre from the starting point. The value of x is:

(A) 4 **(B)** 8 **(C)** 16 **(D)** 32

20. The magnetic field in a plane electromagnetic wave is given by:

$B_y = 2\times10^{-7}\sin\left(\pi\times10^{3}x + 3\pi\times10^{11}t\right)T$. Calculate the wavelength.

(A) $\pi\times10^{-3}m$ **(B)** $2\times10^{-3}m$ **(C)** $2\times10^{3}m$ **(D)** $\pi\times10^{3}m$

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SECTION-2

Section 2 contains 10 Numerical Value Type Questions Out of which ONLY 5 (any) questions have to be attempted. The answer to each question should be **rounded off to the nearest integer**.

1. An rms voltage of 110V is applied across a series circuit having a resistance 11Ω and impedance 22Ω. The power consumed is _____ watt.

2. The frequency of a sonometer wire is 100 Hz. Another sonometer wire of same material but double the length has a frequency 75 Hz. The ratio of tensions in the first wire to the second wire is $\frac{4}{k}$. Find the value of k.

3. The value of wavelength of radiation emitted due to transition of electrons from $n=4$ to $n=2$ state in hydrogen atom will be $\frac{16}{xR}$ (where R is Rydberg constant). Find the value of x.

4. A metal wire carries a charge of 1200 C in 20 minutes. If the area of the cross-section of the wire is $25\ mm^2$ and the material of the metal contains 6.0×10^{22} free electrons/cm^3, the drift velocity of the electrons in the wire is $4.2\times10^{-p}m/s$. Find the value of p.

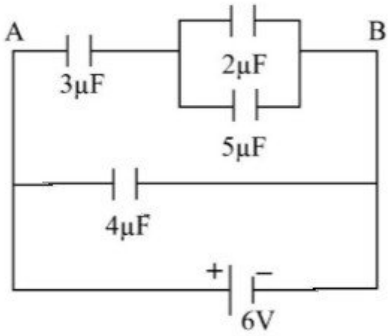
5. A uniform metal rod of $2\ mm^2$ cross-section fixed at both the ends is heated from $0°C$ to $20°C$. The coefficient of linear expansion of the rod is 12×10^{-6} per $°C, Y=10^{11}N/m^2$. The energy stored per unit volume of the rod is _____ Jm^{-3}.

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6. An inclined plane makes an angle of 30° with the horizontal. A solid sphere rolling down on this inclined plane from rest without slipping, has a linear acceleration equal to $\frac{xg}{14}$. Find the value of x.

7. A 1kg block moving with a velocity of $4\ ms^{-1}$ collides with a stationary 2kg block. The lighter block comes to rest after the collision. Find loss of kinetic energy of the system in joule.

8. A circuit is shown in figure. Find the charge (in μC) on the condenser having a capacity of $5\ \mu F$.



9. A spherical surface of radius of curvature R separates air (refractive index 1.0) from glass (refractive index 1.5). The centre of curvature is in glass. A point object P placed in air is found to have a real image Q in the glass. The line PQ cuts the surface at a point O and $PO = OQ$. The distance PO is equal to αR. Find the value of α.

10. The magnetic moment of a current carrying loop is $2.1\times10^{-25}\ Am^2$. The magnetic field at a point on its axis at a distance of $1\mathring{A}$ is $x\times10^{-3}$ weber/m^2. Find the value of x.

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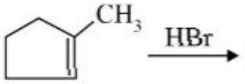
PART II : CHEMISTRY	MARKS: 100

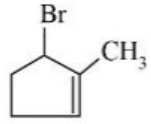
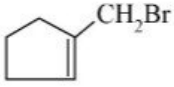
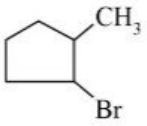
SECTION 1

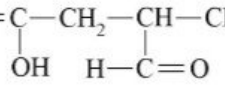
This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. 6g of NaCl was dissolved in water and then treated with excess of silver nitrate solution. If it gives 14 g of AgCl then % purity of NaCl solution would be:

(A) 95% **(B)** 85% **(C)** 75% **(D)** 65%

2.  ?, Major product of the reaction is:

(A)  **(B)**  **(C)**  **(D)** 

3. The correct IUPAC name of the following compound  is:

(A) 3-Formyl-3-oxopropanoic acid **(B)** 3-Formyl-4-oxobutanoic acid

(C) 3, 3-Dioxopropanoic acid **(D)** 3, 3-Dicarbaldehydepropanoic acid

4. 2 mole of an ideal gas at 27°C expands isothermally and reversibly from a volume of 4 litre to 40 litre. The work done (in kJ) by the gas is:

(A) w = –28.72 kJ **(B)** w = –11.488 kJ

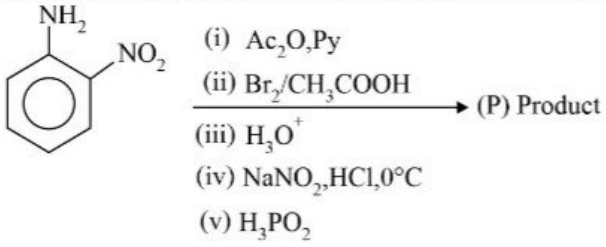
(C) w = –5.736 kJ **(D)** w = –4.988 kJ

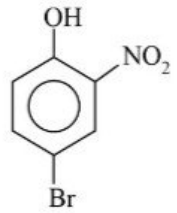
SPACE FOR ROUGH WORK

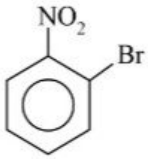
5. Given below are two statements:
Statement I: Phenolphthalein is a weak organic base.
Statement II: The ionized form of Phenolphthalein is colourless.
In the light of the above statement, choose the most appropriate answer from the options given below:
(A) Statement I is correct but Statement II is incorrect
(B) Statement I is incorrect but statement II is correct
(C) Both Statement I and Statement II are incorrect
(D) Both statement I and Statement II are correct

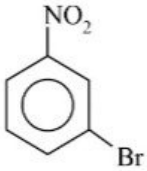
6. Given below are two statements:
Statement I- In redox titration, the $KMnO_4$ acts as self indicator.
Statement II- In titration of weak acid versus strong base, pH at equivalence point is greater than 7
(A) Both statement I and statement II are correct
(B) Statement I is incorrect but Statement II is correct
(C) Statement I is correct but Statement II is incorrect
(D) Both Statement I and statement II are incorrect

7. The product (P) formed from the following multistep reaction is:



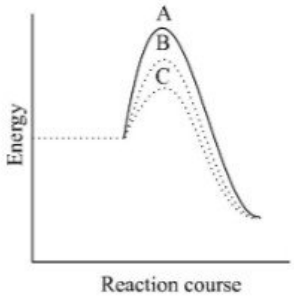
(A) 

(B) 

(C) 

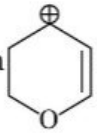
(D) 

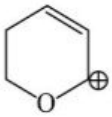
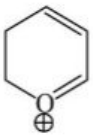
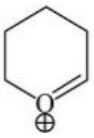
8. A homogeneous catalytic reaction takes place through the three alternative plots A, B and C shown in the given figure. Which one of the following indicates the relative ease with which the reaction can take place?



(A) A > B > C **(B)** C > B > A
(C) A > C > B **(D)** A = B = C

SPACE FOR ROUGH WORK

9. Most stable resonating structure of cation  is:

(A)  **(B)**  **(C)**  **(D)** 

10. Choose the correct code of characteristics for the given order of hybrid orbitals of same atom,

$sp < sp^2 < sp^3$

(i) Electronegativity
(ii) Bond angle between same hybrid orbitals
(iii) Size
(iv) Energy level

(A) (ii), (iii) and (iv)
(B) (iii), (iv)
(C) (ii) and (iv)
(D) (i), (ii), (iii) and (iv)

11. Match List-I and List-II:

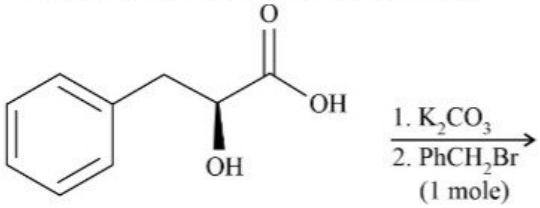
List-I (Complex ion)		**List-II** (Orbital involve in hybridization by central metal)	
(I)	$[NiCl_4]^{2-}$	**(p)**	$3d_{z^2}, 3d_{x^2-y^2}$
(II)	$[Ni(NH_3)_6]^{2+}$	**(q)**	$4d_{z^2}, 4d_{x^2-y^2}$
(III)	$[Ni(CN)_4]^{2-}$	**(r)**	Two 4p-orbitals
(IV)	$[Fe(CN)_6]^{3-}$	**(s)**	No d-orbital

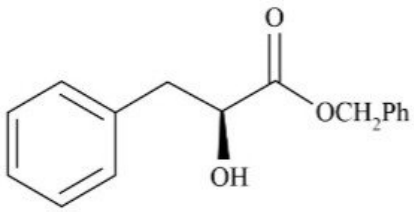
Select **CORRECT** code of your answer.

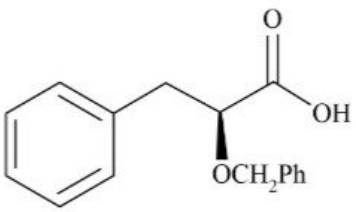
	I	**II**	**III**	**IV**		**I**	**II**	**III**	**IV**
(A)	p	q	r	s	**(B)**	p	r	s	q
(C)	s	r	q	p	**(D)**	s	q	r	p

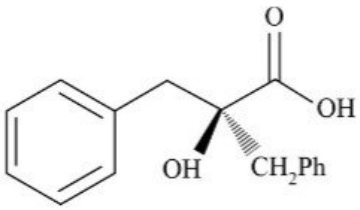
SPACE FOR ROUGH WORK

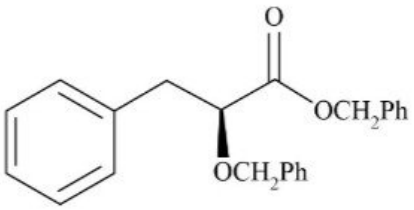
12. The major product in the following reaction



(A)


(B)


(C)


(D)


13. The incorrect statement regarding 15th group hydrides (EH_3). [E = N, P, As, Sb, Bi]

(A) $NH_3 > PH_3 > AsH_3 > SbH_3 > BiH_3$: Thermal stability

(B) $N-H > P-H > As-H > Sb-H > Bi-H$: $E-H$ bond dissociation enthalpy

(C) $NH_3 > PH_3 > AsH_3 > SbH_3 > BiH_3$: Reducing character

(D) $NH_3 > PH_3 > AsH_3 > SbH_3 > BiH_3$: Basicity

14. Given below are two statements: One is labelled as
Assertion A and the other is labelled as Reason R
Assertion (A) : The first ionization enthalpy of aluminium is lower than that of magnesium.

Reason (R): Ionic radius of Al^{3+} is smaller than that of Mg^{2+}.

In the light of above statements, choose the correct answer from the options given below.

(A) Both A and R are true and R is the correct explanation of A.

(B) A is true but R is false.

(C) A is false but R *is* true

(D) Both A and R are true but R is NOT the correct explanation of A

SPACE FOR ROUGH WORK

15. Match the column I with column II and select the correct option:

	Column-I		**Column-II**
(I)	Nucleoside	**(p)**	Sugar + base + phosphoric acid group
(II)	Nucleotide	**(q)**	Cytosine + uracil
(III)	DNA	**(r)**	Sugar + base
(IV)	RNA	**(s)**	Cytosine + thymine

(A) $(I) \rightarrow (r); (II) \rightarrow (p); (III) \rightarrow (s); (IV) \rightarrow (q)$

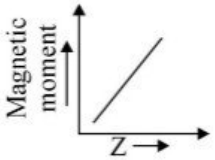
(B) $(I) \rightarrow (p); (II) \rightarrow (r); (III) \rightarrow (s); (IV) \rightarrow (q)$

(C) $(I) \rightarrow (p); (II) \rightarrow (r); (III) \rightarrow (q); (IV) \rightarrow (s)$

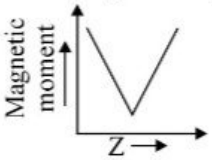
(D) $(I) \rightarrow (r); (II) \rightarrow (p); (III) \rightarrow (q); (IV) \rightarrow (s)$

16. Which of the following graph is the most appropiate representation between atomic number (Z) and magnetic moment of d-block elements while moving from left to right in 4th period?

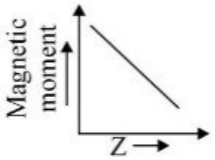
(A)



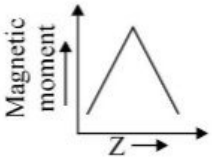
(B)



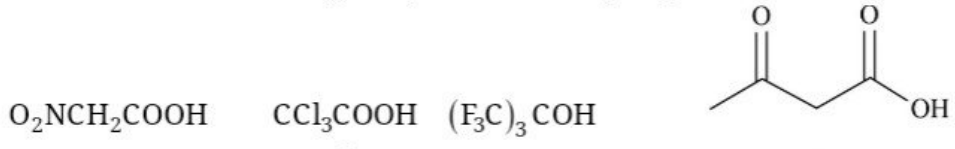
(C)



(D)



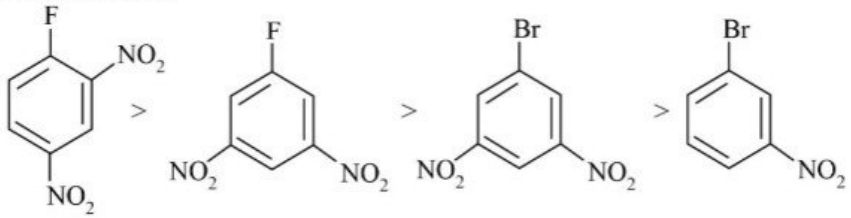
17. The correct order of decreasing acidity for the following compound is:

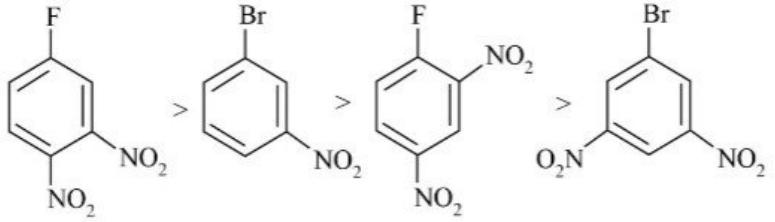


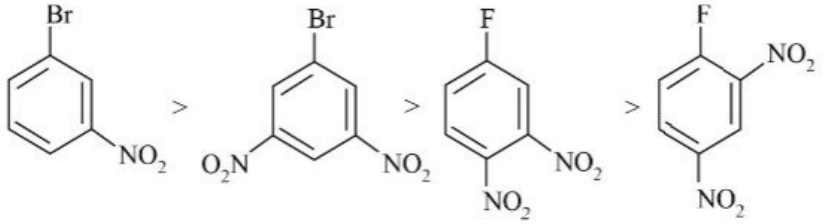
(A) 1 > 3 > 2 > 4 **(B)** 3 > 1 > 2 > 4 **(C)** 2 > 1 > 4 > 3 **(D)** 3 > 1 > 4 > 2

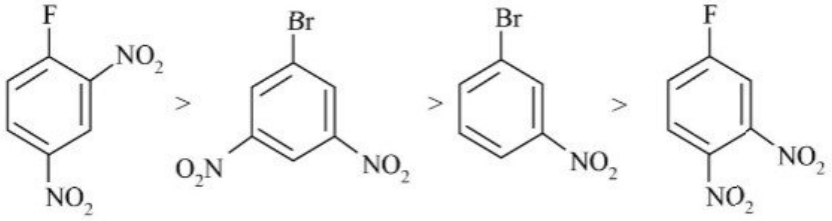
SPACE FOR ROUGH WORK

18. Which of the following correctly ranks the aryl halides in decreasing order of reactivity toward sodium methoxide *in* methanol?

(A) 

(B) 

(C) 

(D) 

19. The common oxidation state of the elements of lanthanide series is:

(A) + 2 **(B)** + 3 **(C)** + 4 **(D)** +1

20. A compound 'X' when treated with chloral in presence of concentrated H_2SO_4 yields 'Y'. 'Y' act as disinfectant. 'X' and 'Y' are respectively:

(A) Chlorobenzene, DDT **(B)** Phenol, DDT

(C) Salicylaldehyde, Phenolphthalein **(D)** Chlorobenzene, Chloropicrin

SPACE FOR ROUGH WORK

SECTION-2

Section 2 contains 10 Numerical Value Type Questions Out of which ONLY 5 (any) questions have to be attempted. The answer to each question should be **rounded off to the nearest integer**.

1. The molar solubility of $Ag_2CO_3\left(K_{sp} = 4\times10^{-13}\right)$ in 0.1 M Na_2CO_3 solution is 10^{-x} then value of x is _____.
2. At 300 K, the half-life of a sample of a gaseous compound initially at 1 atm is 100 sec. When the pressure is 0.5 atm, the half-life is 50 sec. The order of reaction is _____.
3. The number of atomic orbitals from the following having at least one angular node
   7s, 7p, 6s, 8p, 8d.
4. x g sample of an organic compound (Molar mass $80\,g\,mol^{-1}$) is burnt in a Bomb calorimeter. The temperature of calorimeter increase by 4°C. The heat capacity of the system is 1.25 kJ/°C. Calculate the value of *x*. Given molar heat of combustion of the compound at constant volume is $400\ kJ\ mol^{-1}$.
5. Among $SF_3Cl, XeO_2F_2, [ICl_4]^-, [NH_4]^+, [ICl_6]^-, XeO_3, SF_4, [XeF_5]^+$ and $[XeO_6]^{4-}$, the total number of species having sp^3d hybridized central atom is _____.

SPACE FOR ROUGH WORK

6. For a sparingly soluble salt AB_2, concentration of AB_2 in aqueous solution is written as $y\times10^{-3}$M. If saturated aqueous solution shows elevation in boiling point equal to 0.0065 K then find 'y'.

   [Given : Molal elevation constant (K_b) of $H_2O = 0.5K-kg/mol$, solubility product of AB_2 is 2.56×10^{-7}]

7. The sum of oxidation state of the metals in $V_2O_5, Mn_2(CO)_{10}, MnO_2$ is _____.

8. The observed magnetic moment of the complex $\left[Pt(SCN)_2(OX)_2\right]^{x-}$ is zero B.M. The numerical value of x is _____.

9. The number of incorrect statements regarding cathode in a voltaic cell is _____.

   (A) It may gain weight during reaction

   (B) Electrons flow to it through the external circuit

   (C) It is where oxidation occurs

   (D) It receives electrons from ions in solution

10. The maximum number of atoms that can be in same plane in a molecule of B_2H_6 are _____.

SPACE FOR ROUGH WORK

PART III : MATHEMATICS	MARKS: 100

SECTION 1

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. If $\frac{3}{2+\cos\theta+i\sin\theta}=a+ib$, then $(a-2)^2+b^2$ equal to λ, then range of $\frac{1}{\lambda-2\sin\alpha}$ is:

(A) $(0,\infty)$ **(B)** $(-\infty,1]\cup[3,\infty)$

(C) $(-\infty,-1]\cup\left[\frac{1}{3},\infty\right)$ **(D)** $[2,4]\cup[5,\infty)$

2. The reflection of the point P(1, 0, 0) in the line $\frac{x-1}{2}=\frac{y+1}{-3}=\frac{z+10}{8}$ is:

(A) $(3,-4,-2)$ **(B)** $(5,-8,-4)$ **(C)** $(1,-1,-10)$ **(D)** $(2,-3,-8)$

3. If $A=\begin{pmatrix}a & b\\ c & d\end{pmatrix}$ is such that $|A|=0$ and $A^2-(a+d)A+kI=0$, then k is equal to:

(A) $b+c$ **(B)** $a+d$ **(C)** $ab+cd$ **(D)** 0

4. If $\vec{a}$ and $\vec{b}$ are vectors such that $\left|\vec{a}+\vec{b}\right|=\sqrt{29}$ and $\vec{a}\times\left(2\hat{i}+3\hat{j}+4\hat{k}\right)=\left(2\hat{i}+3\hat{j}+4\hat{k}\right)\times\vec{b}$, then a possible value of $\left(\vec{a}+\vec{b}\right)\cdot(-7i+2j+3k)$ is:

(A) 0 **(B)** 3 **(C)** 4 **(D)** 8

5. The coefficient of t^{50} in $\left(1+t^2\right)^{25}\left(1+t^{25}\right)\left(1+t^{40}\right)\left(1+t^{45}\right)\left(1+t^{47}\right)$ is:

(A) $1+{}^{25}C_5$ **(B)** $1+{}^{25}C_5+{}^{25}C_7$

(C) $1+{}^{25}C_7$ **(D)** 1

SPACE FOR ROUGH WORK

6. If common chord of the circle C with centre at (2, 1) and radius r and the circle $x^2+y^2-2x-6y+6=0$ is a diameter of the second circle, then value of r is:

(A) 3 **(B)** 2 **(C)** $\frac{3}{2}$ **(D)** 1

7. If $1, \log_y x, \log_z y, -15\log_x z$ are in A.P., then:

(A) $z^3 = x$ **(B)** $x = y^{-2}$ **(C)** $z^{-2} = y$ **(D)** $x = y$

8. Let $f(x) = \left\{\frac{\log(1+x)^{1+x}}{x^2} - \frac{1}{x}\right\}$. If function f is continuous at $x = 0$ then

$1+2f(0)+3f^2(0)+4f^3(0)....\infty$ is:

(A) 1 **(B)** $\frac{1}{2}$ **(C)** $\frac{1}{4}$ **(D)** 4

9. The number of ways in which we can arrange 4 letters of the word MATHEMATICS is given by:

(A) 136 **(B)** 2454 **(C)** 1680 **(D)** 192

10. Let S denote the sum of all the values of θ $(0 \le \theta \le 2\pi)$ for which the system of equations

$(1+\lambda)x_1 + x_2 + x_3 = 1$

$x_1 + (1+\lambda)x_2 + x_3 = \lambda$

$x_1 + x_2 + (1+\lambda)x_3 = \lambda^2$

is inconsistent where $\tan\theta = \frac{\lambda}{3}$. Then S is:

(A) 3π **(B)** $\frac{5\pi}{4}$ **(C)** $\frac{9\pi}{4}$ **(D)** $\frac{11\pi}{2}$

SPACE FOR ROUGH WORK

11. A line makes angles α, β, γ with the coordinate axes. If $\alpha + \beta = \frac{\pi}{2}$, then $(\cos\alpha + \cos\beta + \cos\gamma)^2$ is equal to:

(A) $1 + \sin 2\alpha$ **(B)** $1 + \cos 2\alpha$ **(C)** $1 - \sin 2\alpha$ **(D)** $1 - \cos 2\alpha$

12. If $4\hat{i} + 7\hat{j} + 8\hat{k}$, $2\hat{i} + 3\hat{j} + 4\hat{k}$ *and* $2\hat{i} + 5\hat{j} + 7\hat{k}$ are the position vectors of the vertices A, B and C, respectively, of triangle ABC, then the position vector of the point where the bisector of angle A meets BC is:

(A) $\frac{2}{3}\left(-6\hat{i} - 8\hat{j} - 6\hat{k}\right)$ **(B)** $\frac{2}{3}\left(6\hat{i} + 8\hat{j} + 6\hat{k}\right)$

(C) $\frac{1}{3}\left(6\hat{i} + 13\hat{j} + 18\hat{k}\right)$ **(D)** $\frac{1}{3}\left(5\hat{j} + 12\hat{k}\right)$

13. In the expansion of $\left(\frac{1}{x^2} - x^3\right)^n, n \in N$, if the sum of the coefficients of x^5 and x^{10} is 0 (which itself are non zero), then term independent of x in the expansion of $\left(x^2 + \frac{1}{x^2} + 2\right)^n$ is:

(A) $\frac{20!}{(10!)^2}$ **(B)** $\frac{1.3.5....15}{2^{15}}$ **(C)** $\frac{30!}{(10!)^2}$ **(D)** $\frac{30!}{(15!)^2}$

14. The relation R given by $\{(x, y) : x^2 - 3xy + 2y^2 = 0, \forall x, y \in R\}$ is:

(A) reflexive but not symmetric **(B)** symmetric but not transitive

(C) symmetric and transitive **(D)** an equivalence relation

15. Three numbers are chosen at random from the set of integers 1, 2, 3, …., 39. Then probability that number are in AP is:

(A) $\frac{1}{247}$ **(B)** $\frac{2}{247}$ **(C)** $\frac{19}{481}$ **(D)** $\frac{5}{741}$

SPACE FOR ROUGH WORK

16. $\int a^x\left(\ln x+\ln a.\ln\left(\frac{x}{e}\right)^x\right)dx=$

(A) $a^x\ln\left(\frac{e}{x}\right)^{2x}+C$ **(B)** $a^x\ln\left(\frac{x}{e}\right)^x+C$

(C) $a^x+\ln\left(\frac{x}{e}\right)^x+C$ **(D)** $a^x\ln(ex)^x+C$

17. The value of the expression $\left(2\sin^2 91°-1\right)\left(2\sin^2 92°-1\right)....\left(2\sin^2 180°-1\right)$ is equal to:

(A) 0 **(B)** 1 **(C)** 2^{90} **(D)** $2^{90}-90$

18. PSQ is a focal chord of a parabola whose focus is S and vertex is A. PA and QA are produced to meet the directrix in R and T, respectively. Then $\angle RST=$

(A) 30° **(B)** 90° **(C)** 60° **(D)** 45°

19. Consider the triangles with vertices A(2, 1), B(0, 0) and C(t, 4), t = [0, 4]. If the maximum and the minimum perimeters of such triangles are obtained at $t=\alpha$ *and* $t=\beta$ respectively then $6\alpha+21\beta$ is equal to:

(A) 27 **(B)** 33 **(C)** 39 **(D)** 48

20. A number equal to 2 times the mean and with a frequency equal to k is inserted in a data having n observations. If the new mean is $\frac{4}{3}$ times the old mean, then the value of $\frac{k}{n}$ is:

(A) 1 **(B)** 2 **(C)** $\frac{1}{3}$ **(D)** $\frac{1}{2}$

SPACE FOR ROUGH WORK

SECTION-2

Section 2 contains 10 Numerical Value Type Questions Out of which ONLY 5 (any) questions have to be attempted. The answer to each question should be **rounded off to the nearest integer**.

1. Let $f(x)=\frac{9^x}{9^x+3}$ then the value of $f\left(\frac{1}{1996}\right)+f\left(\frac{2}{1996}\right)+f\left(\frac{3}{1996}\right)+\ldots+f\left(\frac{1995}{1996}\right)$ is k then $\left|k-\frac{1}{2}\right|$ will be _____.

2. If x_1, x_2, x_3 are such that $x_1+x_2+x_3=2$

$x_1^2+x_2^2+x_3^2=6$

$x_1^3+x_2^3+x_3^3=8$

then value of $\left|(x_2-x_3)(x_3-x_1)(x_1-x_2)\right|$ is _____.

3. The perpendicular bisector of a line segment with end point (1, 2, 6) and (–3, 6, 2) passes through (–6, 2, 4) and has the equation of the form $\frac{x+6}{l}=\frac{y-2}{m}=\frac{z-4}{n}$ (where l, m, n are integers, l is a prime number and $l>0$), then the value of $(l+m+n)-lmn$ equals to _____.

4. The domain of definition of the function $f(x)=\sqrt{\log_{(|x|-1)}\left(x^2+4x+4\right)}$, is $(-\infty, a]\cup(b,c)\cup(d,\infty)$ then $|a+b+c+d|=$_____.

5. If the area bounded by $y=x^2$ *and* $y=\frac{2}{1+x^2}$ is $\left(K_1\pi-\frac{K_2}{3}\right)$ *sq.* units (where $K_1, K_2 \in Z$), then the value of (K_1+K_2) is equal to _____.

SPACE FOR ROUGH WORK

6. In an increasing sequence of four positive integers, the first 3 terms are in A.P., the last 3 terms are in G.P. and the fourth term exceed the first term by 30, then the common difference of A.P. lying in interval [1, 9] is _____.

7. Let $\frac{dy}{dx}=\frac{y\phi'(x)-y^2}{\phi(x)}$, where $\phi(x)$ is a specified function satisfying $\phi(1)=1, \phi(4)=1296$. If $y(1)=1$ then $\frac{1}{81}y(4)$ is equal to _____.

8. If $\int_{200\pi}^{0}[\sin x+\cos x]dx=A\pi$ and $[x]$ denotes the greatest integer function than A is equal to _____.

9. If the parabola $x^2=4y$ and the circle $x^2+(y-7)^2=r^2$ have maximum number of common chords, then least value of $[r]$ ([x] denotes the greatest integer $\le x$).

10. If the function $f(x)=x^3+e^{\frac{x}{2}}$ *and* $g(x)=f^{-1}(x)$, then the value of $g'(1)$ is _____.

SPACE FOR ROUGH WORK

End of 2024-Mock JEE Main - 2

Solutions to Mock JEE MAIN – 2 | JEE - 2024

Physics

SINGLE CHOICE

1.(B) Since, potential V is same for all points of the sphere. Therefore, we can calculate its value at the centre of the sphere.

$$\therefore \qquad V = \frac{1}{4\pi\varepsilon_0}.\frac{q}{r} + V'$$

V' = potential at centre due to induced charges = 0 (because net induced charge will be zero)

$$\therefore \qquad V = \frac{1}{4\pi\varepsilon_0}.\frac{q}{r}$$

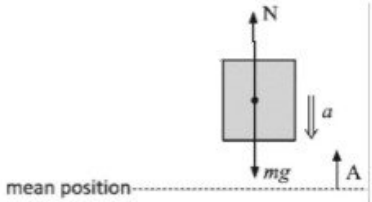
2.(A) $A_R = \sqrt{\left(\sqrt{3}\right)^2 + 1^2} = 2 \qquad \phi = \tan^{-1}\left(\frac{\sqrt{3}}{1}\right) = \frac{\pi}{3};$

So equation of SHM is : $\qquad y = 2\sin\left(\omega t + \frac{\pi}{3}\right)$

Maximum chance of break off is at extreme position. $mg - N = m\omega^2 A$

For breakoff $N = 0 \qquad \Rightarrow \qquad \omega = \sqrt{\frac{g}{A}} = \sqrt{\frac{g}{2}}$

Also for $y = A = 2 \qquad \Rightarrow \qquad \omega t + \frac{\pi}{3} = \frac{\pi}{2} \quad \Rightarrow \quad t = \frac{\pi}{6\omega} = \frac{\pi}{6}\sqrt{\frac{2}{g}}$



3.(B) $V = IR$

$$\Rightarrow \qquad R = \frac{V}{I} = \frac{U}{qI} = \frac{U}{I^2 t} = \frac{\left(ML^2T^{-2}\right)}{\left[A^2\right][T]} = \left[ML^2T^{-3}A^{-2}\right]$$

$$\Rightarrow \qquad C = \frac{Q}{V} = \frac{It}{U/q} = \frac{I^2t^2}{U} = \frac{\left[A^2\right]\left[T^2\right]}{\left[ML^2T^{-2}\right]} = \left[M^{-1}L^{-2}T^4A^2\right]$$

$$\Rightarrow \qquad F = iBl$$

$$B = \frac{F}{il} = \frac{\left[MLT^{-2}\right]}{[A][L]} = \left[MT^{-2}A^{-1}\right]$$

$$\Rightarrow \qquad U - \frac{1}{2}Li^2$$

$$L = \frac{U}{i^2} = \frac{\left[ML^2T^{-2}\right]}{\left[A^2\right]} = \left[ML^2T^{-2}A^{-2}\right]$$

4.(B) $I = \frac{100\times10^{-3}}{0.5}A = 0.2A$

$$R = \frac{1}{0.2}\Omega = 5\Omega.$$

5.(B) Comparing the given equation with $y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta}$, we get $\tan\theta = \sqrt{3}$ and $u = 2\,\text{m/s}$.

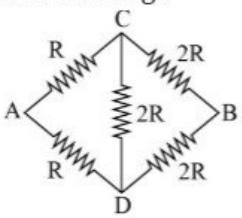
6.(A) $\frac{1}{\lambda} = R\left[1 - \frac{1}{n^2}\right]$

or $\frac{1}{\lambda R} = 1 - \frac{1}{n^2} \Rightarrow \frac{1}{n^2} = 1 - \frac{1}{\lambda R}$

or $\frac{1}{n^2} = \frac{\lambda R - 1}{\lambda R} \Rightarrow n^2 = \frac{\lambda R}{\lambda R - 1}$

or $n = \left(\frac{\lambda R}{\lambda R - 1}\right)^{1/2}$.

7.(C) It is wheat stone bridge



8.(A) Equating pressures, $\frac{F}{\pi R^2} = \frac{1200}{\pi[20R]^2} \Rightarrow F = 3kgf$

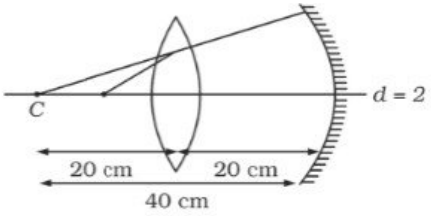
9.(C) The silvered lens can be replaced by a mirror of focal length given as

$\frac{1}{F_M} = \frac{1}{f_m} - \frac{2}{f_1}, \frac{1}{F_M} = 0 - \frac{2}{40}\ F_M = -20$

For lens $v = \frac{uf}{u + f'}$ $v = \frac{-10 \times 20}{-10 + 20} = -20$

So, this position has to be centre of curvature of mirror in order for the ray to retrace its path so,

$d = 40 - 20 = 20cm$



10.(B) $T_1 = 27 + 273 = 300K$

$T_2 = 127 + 273 = 400K$

$\frac{E_2}{E_1} = \frac{T_2}{T_1} \Rightarrow E_2 = \frac{4}{3} \times 6.2 \times 10^{-21} J = 8.27 \times 10^{-21} J$

11.(D) $\frac{a_m}{g} = \frac{GM_e / R_m^2}{GM_e / R_e^2}$ or $\frac{a_m}{g} = \frac{R_e^2}{R_m^2}$ or $a_m = \left[\frac{R_e}{R_m}\right]^2 g$

12.(B) $U = \frac{\frac{3}{2}RT + \frac{6}{2}RT}{2} = \frac{9}{4}RT; C_v = \frac{dU}{dT} = \frac{9}{4}R$

$C_p = C_v + R = \frac{13}{4}R \Rightarrow \frac{C_p}{C_v} = \frac{13}{9} = 1.44$

13.(C) When a body is at highest position motion under gravity, then at that time velocity is zero and acceleration is g.

14.(C) $\lambda_{min} = \dfrac{hc}{eV}$

15.(B) $\beta = \dfrac{3}{10} = 0.3cm$

$\beta = \dfrac{D\lambda}{d}$

$\therefore\ 0.3 = \dfrac{300\times 5100\times 10^{-8}}{d}$ or $d = \dfrac{300\times 5100\times 10^{-8}}{0.3}cm$

$= 51\times 10^{-3}cm = 51\times 10^{-2}mm = 0.51\,mm$

16.(A) $\vec{u} = 10\hat{i}\ m/s$

$\vec{a} = 2\,\hat{j}\ m/s^1$

$\vec{v} = \vec{u} + \vec{a}t$

$\vec{v} = 10\hat{i} + 2\,\hat{j}\times 5 = \left(10\hat{i} + 10\,\hat{j}\right)m/s$

$v = 10\sqrt{2}m/s = 14m/s$

17.(C) $eV_s = h\nu - \phi_0$

$eV_s' = h\nu' - \phi_0$

$e\left(V_s' - V_s\right) = h\nu' - h\nu = \left(\dfrac{12375}{3600} - \dfrac{12375}{4000}\right)eV$

$\therefore\quad V_s' - V_s = 3.44 - 3.09 = 0.35V$

18.(B) The emf induced across the rod AB is

$e = Bv_\perp l$

Here, $v_\perp = v\sin 30^\circ$ = component of velocity perpendicular to length

$\therefore\quad e = Bvl\sin 30^\circ$

$= (2)(4)(1)\left(\dfrac{1}{2}\right) = 4V$

The free electrons of the rod shift towards right due to the force $q\left(\vec{v}\times\vec{B}\right)$. Thus, the left side of the rod is at higher potential.

or $V_A - V_B = 4V$

19.(C) Force on block = $20\times 2N = 40N$

Frictional force on block = $0.15\times 20\times 10N = 30N$

Net force = 10 N

Acceleration = $\dfrac{10N}{20kg} = \dfrac{1}{2}ms^{-2}$

Now, $4 = 0\times t + \dfrac{1}{2}\times\dfrac{1}{2}t^2\ or\ t = 4s$

Let us calculate the distance travelled by the truck.

$x = 0\times 4 + \dfrac{1}{2}\times 2\times 4\times 4 = 16m$

20.(B) Coefficient of x is angular wave number k of $\frac{2\pi}{\lambda}$.

Thus, $k=\frac{2\pi}{\lambda}=\pi\times10^{3}$

$\therefore \quad \lambda=2\times10^{-3}m$

NUMERICAL TYPE

1.(275) $P=E_vI_v\cos\phi$

$P=E_v\frac{E_v}{Z}\frac{R}{Z}$

or $\quad P=\frac{E_v^2R}{Z^2}=\frac{110\times110\times11}{22\times22}W=275\,W.$

2.(9) $v=\frac{1}{2l}\sqrt{\frac{T}{m}}\ or\ v\propto\frac{\sqrt{T}}{l}\ or\ \sqrt{T}\propto vl$

$\therefore\ \sqrt{\frac{T_1}{T_2}}=\frac{100l}{75(2l)}=\frac{2}{3}\ or\ \frac{T_1}{T_2}=\frac{4}{9}$

3.(3) $\frac{1}{\lambda}=R\left[\frac{1}{n_1^2}-\frac{1}{n_2^2}\right]\ or\frac{1}{\lambda}=R\left[\frac{1}{4}-\frac{1}{16}\right]$

$\frac{1}{\lambda}=R\left[\frac{4-1}{16}\right]\ or\lambda=\frac{16}{3R}$

4.(6) Given, $q=1200C, t=20$ minute, A = 25 $mm^2=25\times10^{-6}m^2$

$n=6.0\times10^{22}$ electrons/$cm^3=6.0\times10^{22}\times10^{6}/m^3$

Drift velocity $v_d=?$

We know that, $I=\frac{q}{t}=\frac{1200}{20\times60}=1A$

Drift velocity, $v_d=\frac{I}{neA}$

$v_d=\frac{1}{6\times10^{22}\times10^{6}\times25\times10^{-6}\times1.6\times10^{-19}}$

$v_d=\frac{4.16\times10^{-3}}{10^{3}}$

$v_d=4.2\times10^{-6}m/s$

5.(2880) Energy per unit volume $=\frac{1}{2}\times$ stress $\times$ strain $=\frac{1}{2}(Y\alpha t)(\alpha t)=\frac{1}{2}Y\alpha^2t^2$

$=\frac{10^{11}\times144\times10^{-12}\times400}{2}J\ m^{-3}$

$=288\times10J\ m^{-3}=2880\,J\ m^{-3}$

6.(5) $a=\dfrac{g\sin\theta}{1+\dfrac{K^2}{R^2}}=\dfrac{g\sin30^\circ}{1+\dfrac{2}{5}}$ or $a=\dfrac{5}{7}g\times\dfrac{1}{2}=\dfrac{5g}{14}$

7.(4) Applying conservation of momentum, $1\times4=2\times v$ *or* $v=2ms^{-1}$

Loss of kinetic energy $=\dfrac{1}{2}\times1\times4\times4-\dfrac{1}{2}\times2\times2\times2=(8-4)J=4J$

8.(9) Potential difference between A and B = 6 volts. The condensers $2\,\mu F$ *and* $5\mu F$ are in parallel. Their effective capacitance, C = 2 + 5 = 7 μF.

The capacitance between A and B is given by $C'=\dfrac{C\times3}{C+3}=\dfrac{7\times3}{7+3}=\dfrac{21}{10}\mu F$

Total charge $Q=CV=\dfrac{21}{10}\times6=\dfrac{63}{5}\mu C$

Total potential difference across $3\mu F$ is

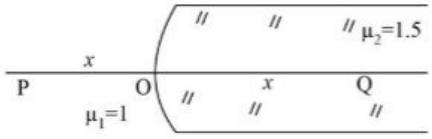
$V_1=\dfrac{Q}{3}=\dfrac{63}{5}\times\dfrac{1}{3}=\dfrac{21}{5}$ volts

$V_2=6-\dfrac{21}{5}=\dfrac{9}{5}V$

So, the charge on $5\,\mu F$ condenser is

$Q=5\times V_2=5\times\dfrac{9}{5}=9\mu C$

9.(5) $\dfrac{\mu_1}{-u}+\dfrac{\mu_2}{v}=\dfrac{\mu_2-\mu_1}{R}$



$\dfrac{1}{-(-x)}+\dfrac{1.5}{x}=\dfrac{1.5-1}{R}$

$\dfrac{1}{x}+\dfrac{3}{2x}=\dfrac{1}{2R}$

$\dfrac{5}{2x}=\dfrac{1}{2R}$ *or* $x=5R$

10.(42) Find at a point x from the centre of a current carrying loop on the axis is

$B=\dfrac{\mu_0}{4\pi}.\dfrac{2M}{x^3}=\dfrac{10^{-7}\times2\times2.1\times10^{-25}}{\left(10^{-10}\right)^3}$

$=4.2\times10^{-32}\times10^{30}=4.2\times10^{-2}W$

Chemistry

SINGLE CHOICE

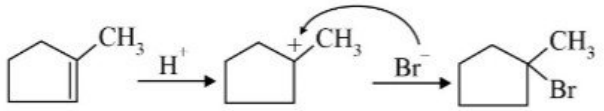
1.(A) The reaction that takes place is $NaCl + AgNO_3 \longrightarrow AgCl\downarrow + NaNO_3$

$\because$ 143.5g of AgCl is produced from 58.5g NaCl

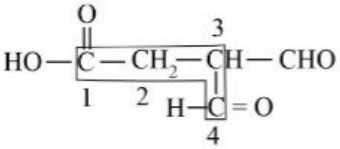
$\therefore$ 14g of AgCl will be produced from $\frac{58.5\times14}{143.5} = 5.70g$ NaCl

5.70g is the amount of NaCl in common salt; % purity = $\frac{5.70}{6}\times100 = 95\%$

2.(D)



3.(B)



3 – formyl – 4 – oxo butanoic acid

4.(B) $W = -nRT\ln\frac{V_2}{V_1}$

$W = -2\times8.314\times10^{-3}\times300\ln\frac{40}{4}$,

$W = -11.488kJ$

5.(C) Phenolphthalein is a weak organic acid and represented by Ph-H

$PhH \rightleftharpoons Ph^{(-)} + H^{(+)}$

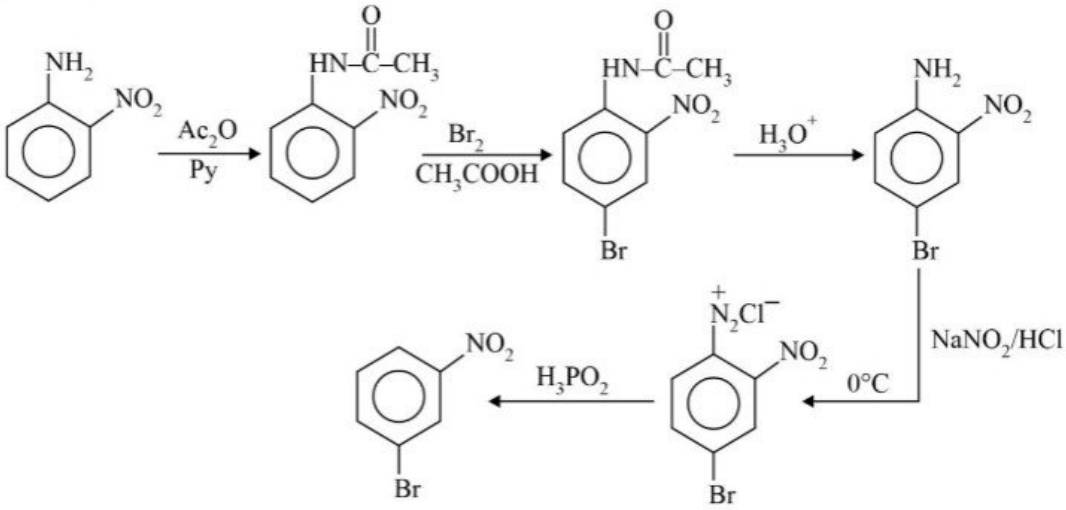
Colorless pink colour

Statement I- False

Statement II- False

6.(A) In redox titration, $KMnO_4$ act as self indicator and because of hydrolysis of anion, pH at equivalence point is greater than 7.

7.(C)



8.(B) More activation energy, slow reaction rate.

9.(B) According to Lewis, structure with complete octet of most of the 2nd period element is the most stable.

10.(B) More percentage s-character means more electronegativity, more bond angle, less size and less energy.

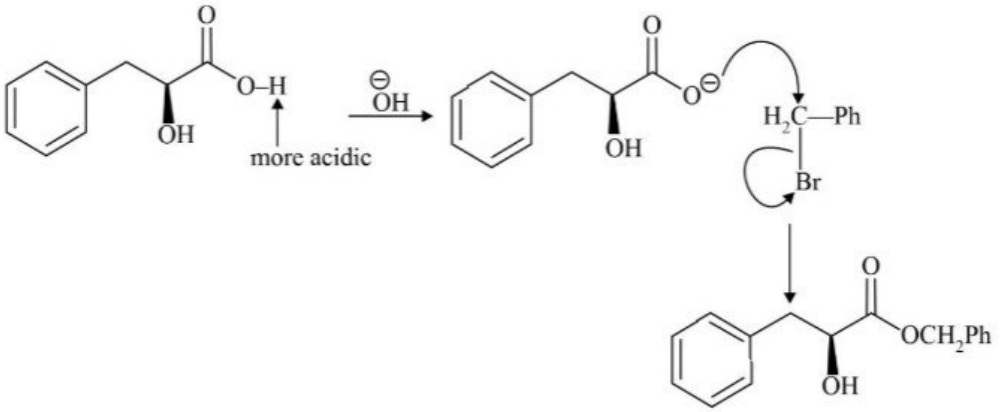
11.(D) $[NiCl_4]^{2-} = sp^3\left(4s + 4p_x + 4p_y + 4p_z\right)$

$\left[Ni(NH_3)_6\right]^{2+} = sp^3d^2\left(4s + 4p_x + 4p_y + 4p_z + 4d_{x^2-y^2} + 4d_{z^2}\right)$

$\left[Ni(CN)_4\right]^{2-} = dsp^2\left(3d_{x^2-y^2} + 4s + 4p_x + 4p_y\right)$

$\left[Fe(CN)_6\right]^{3-} = d^2sp^3\left(3d_{x^2-y^2} + 3d_{z^2} + 4s + 4p_x + 4p_y + 4p_z\right)$

12.(A)



13.(C) Correct reducing character is $NH_3 < PH_3 < AsH_3 < SbH_3 < BiH_3$.

14.(D) IE $\rightarrow$ Mg > Al

$1s^2\,2s^2\,2p^6\,3s^2$ $\qquad$ $1s^2\,2s^2\,2p^6\,3s^2\,3p^1$

Due to stable configuration of Mg.

Size : $Mg^{2+} > Al^{3+}$

More positive charge means more Z_{eff} so size decreases.

15.(A)

16.(D) Magnetic moment $= \sqrt{n(n+2)}BM$

n : Number of unpaired e^-

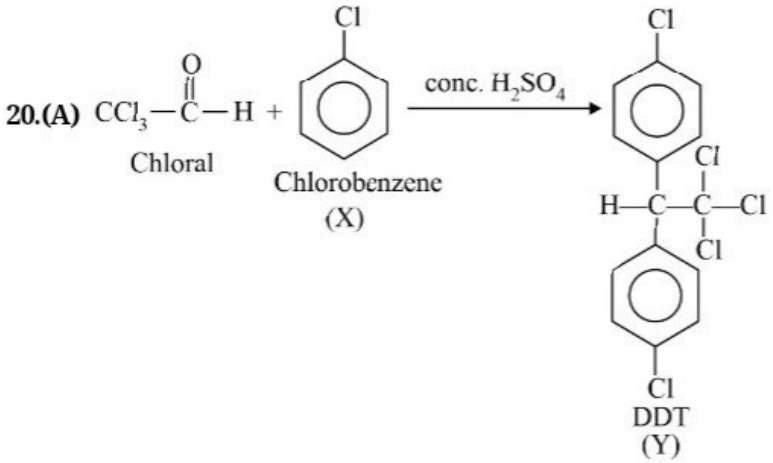
As atomic number increases in d-block element number of unpaired e^- first increases upto 6 then decreases.

17.(C) Carboxylic acids are more acidic than alcohol. CCl_3 at $\alpha-$ position has greater E.W. effect than NO_2 at $\beta-$ position. Carbonyl F.G at $\beta-$ position has lesser E.W. effect than NO_2 group.

18.(A) More the electron withdrawing groups at ortho and para position more is the reactivity.

19.(B) For f-block elements common O.S. is +3.

20.(A)



NUMERICAL TYPE

1.(6) Let solubility of Ag_2CO_3 in presence of Na_2CO_3 is S

$$Ag_2CO_3(s) \rightleftharpoons \underset{2S}{2Ag^+(aq)} + \underset{(S+0.1)}{CO_3^{2-}(aq)}$$

$$K_{sp} = \left[Ag^+\right]^2\left[CO_3^{2-}\right]$$

$$\Rightarrow 4\times10^{-13} = (2S)^2 \times 0.1$$

$$\Rightarrow S = 10^{-6} \Rightarrow x = 6$$

2.(0) For zero order reaction, $t_{1/2} \propto [\text{initial conc.}]^1$

3.(3) Angular node = ℓ

$7s \Rightarrow A.N = 0$

$7p \Rightarrow A.N = 1$

$6s \Rightarrow A.N = 0$

$8p \Rightarrow A.N = 1$

$8d \Rightarrow A.N = 2$

4.(1) $Q_v = 1.25\times4 = 5$

$$Q_v = 5 = \frac{x}{80}\times400 \Rightarrow x = 1$$

5.(3) SF_3Cl, XeO_2F_2, SF_4

6.(1) $\underset{S-x}{AB_2(aq)} \rightleftharpoons \underset{x}{A^{2+}(aq)} + \underset{2x}{2B^-(aq)}$

$$K_{sp} = 2.56\times10^{-7} = 4x^3; x = 4\times10^{-3}M$$

$$\Delta T_b = K_b(S+2x)$$

$$6.5\times10^{-3} = 0.5\left(S+2\times4\times10^{-3}\right); S = 5\times10^{-3}$$

Concentration of $AB_2(aq) = S - x = 5\times10^{-3} - 4\times10^{-3} = 1\times10^{-3}M$

7.(9) Sum of oxidation state = 5 + 0 + 4 = 9

8.(2) $\left[Pt(SCN)_2(OX)_2\right]^{X-}$

Number of unpaired electron = 0

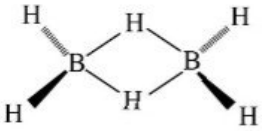
So, Pt must be in +4 oxidation state $\left[Pt(SCN)_2(OX)_2\right]^{X-}$

$\Rightarrow 4+(-6)=-x$

$\Rightarrow x=2$

9.(2) At cathode Reduction takes place so it may gain weight during reaction when metal is deposited after reduction. (A and B statements are correct)

10.(6)



Mathematics

SINGLE CHOICE

1.(C) $(a-2)+ib=-\dfrac{(1+2\cos\theta)+2i\sin\theta}{(2+\cos\theta)+i\sin\theta}$

$\Rightarrow\ (a-2)^2+b^2=\dfrac{(1+2\cos\theta)^2+4\sin^2\theta}{(2+\cos\theta)^2+\sin^2\theta}=\dfrac{5+4\cos\theta}{5+4\cos\theta}=1$

$\lambda=1$

$\therefore\ -1\le 1-2\sin\alpha\le 3$

Range $(-\infty,-1]\cup\left[\dfrac{1}{3},\infty\right)$

2.(B) Coordinates of any point Q on the given line are $(2r+1,-3r-1,8r-10)$.

So the direction ratios of PQ are $2r,-3r-1,8r-10$.

Now PQ is perpendicular to the given line.

if $2(2r)-3(-3r-1)+8(8r-10)=0$

$\Rightarrow\quad 77r-77=0\Rightarrow r=1$

and the coordinates of *Q, the foot of* the perpendicular from P on the line are $(3,-4,-2)$.

Let $R(a,b,c)$ be the reflection of P in the given lines then Q is the mid-point of PR

$\Rightarrow\quad \dfrac{a+1}{2}=3,\dfrac{b}{2}=-4,\dfrac{c}{2}=-2$

$\Rightarrow\quad a=5,b=-8,c=-4$

and the coordinate of the required point are (5, –8, –4).

3.(D) As $|A|=0$, we get $ad-bc=0$

Also, $A^2-(a+d)A=A\left[A-(a+d)I\right]$

$=\begin{pmatrix}a & b\\ c & d\end{pmatrix}\begin{pmatrix}-d & b\\ c & -a\end{pmatrix}=\begin{pmatrix}-ad+bc & ab-ba\\ -cd+cd & bc-ad\end{pmatrix}=\begin{pmatrix}0 & 0\\ 0 & 0\end{pmatrix}$

Thus, $-kI=0\Rightarrow k=0$

4.(C) $\vec{a}\times(2i+3j+4k)=(2i+3j+4k)\times\vec{b}=-\vec{b}\times(2i+3j+4k)$

$\Rightarrow\ (\vec{a}+\vec{b})\times(2i+3j+4k)=0$

$\Rightarrow\ (\vec{a}+\vec{b})$ is parallel to $(2i+3j+4k)=0$

$\Rightarrow(\vec{a}+\vec{b})=\alpha(2i+3j+4k)$ for some $\alpha\in R$

$\Rightarrow|\vec{a}+\vec{b}|=|\alpha|\sqrt{4+9+16}$

$\Rightarrow\ \alpha=\pm1$

Also, $(\vec{a}+\vec{b})\cdot(-7\hat{i}+2\hat{j}+3\hat{k})$

$= \alpha\left(2\hat{i}+3\hat{k}+4\hat{k}\right)\cdot\left(-7\hat{i}+2\hat{j}+3\hat{k}\right)$

$= \alpha(-14+6+12) = 4\alpha$

Taking $\alpha = 1$, we get one of the possible values as 4.

5.(A) As we are interested in coefficient of t^{50}, we shall ignore all the term with exponent more than 50. Thus, we can write given expression as

$\left(1+{}^{25}C_1\,t^2+\ldots+{}^{25}C_{25}\,t^{50}\right)\times\left(1+t^{25}+t^{40}+t^{45}+t^{47}\right)$

As all the terms in the first expression have even exponent we can ignore t^{25}, t^{45} *and* t^{47} too. Thus, coefficient of t^{50} in (1) $= {}^{25}C_{25}+{}^{25}C_5 = 1+{}^{25}C_5$

6.(A) Equation of the circle C is $(x-2)^2+(y-1)^2 = r^2$

$\Rightarrow x^2+y^2-4x-2y+5-r^2 = 0$

Equation of the common chord is $\left(x^2+y^2-4x-2y+5-r^2\right)-\left(x^2+y^2-2x-6y+6\right)=0$

$\Rightarrow 2x-4y+r^2+1=0$

If it is a diameter of the second circle, it passes through the centre (1, 3) of the circle.

So $2-4\times3+r^2+1=0\Rightarrow r^2=9\Rightarrow r=3$

7.(A) Let d be the common difference. Then $\log_y x = 1+d \Rightarrow x = y^{1+d}$

$\log_z y = 1+2d \Rightarrow y = z^{1+2d}$

and $-15\log_x z = 1+3d \Rightarrow z = x^{-(1+3d)/15}$

$x = y^{1+d} = z^{(1+2d)(1+d)}$

$= x^{-(1+d)(1+2d)(1+3d)/15}$

$\Rightarrow (1+d)(1+2d)(1+3d) = -15$

$\Rightarrow 6d^3+11d^2+6d+16=0$

$\Rightarrow (d+2)\left(6d^2-d+8\right)=0\Rightarrow d=-2$

[Note that $6d^2-d+8=0$ has complex roots]

$\therefore \quad x = y^{1+d} = y^{-1}, y = z^{1-4} = z^{-3}$ and $x = \left(z^{-3}\right)^{-1} = z^3$.

8.(D) $f(0) = \lim\limits_{x\to0} f(x)$

$= \lim\limits_{x\to0}\dfrac{(1+x)\log(1+x)-x}{x^2}\ \left(\dfrac{0}{0}\,form\right)$

$= \lim\limits_{x\to0}\dfrac{\log(1+x)+1-1}{2x}$

$$=\frac{1}{2}\lim_{x\to 0}\frac{\log(1+x)}{x}=\frac{1}{2}$$

$S_\infty=\frac{a}{1-r}+\frac{dr}{(1-r)^2}$ (As given expression is an AGP)

$$=\frac{1}{1/2}+\frac{1/2}{1/4}=4$$

9.(B) (M, M), (A, A), (T, T), E, H, I, C, S

We can arrange 4 distinct letters in ${}^8P_4$ ways, 2 identical and 2 distinct in $\left({}^3C_1\right)\left({}^7C_2\right)\frac{4!}{2!}$ ways and 2 pairs of identical letters in $\left({}^3C_2\right)\frac{4!}{2!2!}$ ways

10.(D) If A denotes the coefficient matrix, then $|A|=\lambda^2(\lambda+3)$

For $\lambda=0$ *and* $\lambda=-3$, the system is inconsistent.

$\tan\theta=0$ $\qquad$ $\tan\theta=-1$

$\Rightarrow\ \theta=0,\pi,2\pi$ $\qquad$ $\theta=\frac{3\pi}{4},\frac{7\pi}{4}$

$S=\frac{11\pi}{2}$

11.(A) We have $\cos^2\alpha+\cos^2\beta+\cos^2\gamma=1$

$\Rightarrow\ \cos^2\alpha+\cos^2\left(\frac{\pi}{2}-\alpha\right)+\cos^2\gamma=1$

$\Rightarrow\ \cos^2\alpha+\sin^2\alpha+\cos^2\gamma=1$

$\Rightarrow\ \cos^2\gamma=0\Rightarrow\cos\gamma=0$

So $(\cos\alpha+\cos\beta+\cos\gamma)^2=(\cos\alpha+\sin\alpha)^2=1+2\sin\alpha\cos\alpha$

$=1+\sin2\alpha$

12.(C) Suppose the bisector of angle A meets BC at D. Then AD divides BC in the ratio AB : AC. So, P.V. of D is given by

$$\frac{\left|\overrightarrow{AB}\right|\left(2\hat{i}+5\hat{j}+7\hat{k}\right)+\left|\overrightarrow{AC}\right|\left(2\hat{i}+3\hat{j}+4\hat{k}\right)}{\left|\overrightarrow{AB}\right|+\left|\overrightarrow{AC}\right|}$$

But $\overrightarrow{AB}=-2\hat{i}-4\hat{j}-4\hat{k}$

and $\overrightarrow{AC}=-2\hat{i}-2\hat{j}-\hat{k}$

$\Rightarrow\quad \left|\overrightarrow{AB}\right|=6$ *and* $\left|\overrightarrow{AC}\right|=3$

Therefore, P.V. of D is given by

$$\frac{6\left(2\hat{i}+5\hat{j}+7\hat{k}\right)+3\left(2\hat{i}+3\hat{j}+4\hat{k}\right)}{6+3}=\frac{1}{3}\left(6\hat{i}+13\hat{j}+18\hat{k}\right)$$

13.(D) $\left(\frac{1}{x^2}-x^3\right)^n$

General term $T_{r+1}=\frac{n!}{r!(n-r)!}(-1)^r x^{5r-2n}$

If $5r-2n=5$, then $5r=2n+5$ or $r=\frac{2n}{5}+1$

If $5r-2n=10$, then $5r=2n+10$ or $r=\frac{2n}{5}+2$

x^5 *and* x^{10} terms occurs if $n=5k$

Given that sum of x^5 *and* x^{10} is zero.

$$\Rightarrow \quad \frac{5k!}{(2k+1)!(3k-1)!}-\frac{5k!}{(2k+2)!(3k-2)!}=0$$

or $$\frac{1}{3k-1}-\frac{1}{2k+2}=0$$

or $k=3\Rightarrow n=15$

$$\left(x^2+\frac{1}{x^2}+2\right)^{15}$$

$$\left(x+\frac{1}{x}\right)^{30}$$

Term independent of x is 16th term, $T_{16}={}^{30}C_{15}=\frac{30!}{(15!)^2}$

14.(A) Reflexive but not symmetric

$\because \quad x^2-3xy+2y^2=0$

$\Rightarrow \quad x^2-xy-2xy+2y^2=0$

$\Rightarrow \quad x(x-y)-2y(x-y)=0$

$\Rightarrow \quad (x-2y)(x-y)=0$

$\Rightarrow \quad x=y \text{ or } x=2y$

Now, $\because$ in R all ordered pairs (x, x) are present

$\therefore$ It is reflexive

Now, $(4,2)\in R$ as 4 = 2 (2)

but (2, 4) $\notin$ R as $2\neq 2(4)$

$\therefore$ It is not symmetric

Also (4, 2) and (2, 1) $\in R$ but $(4,1)\notin R$

$\therefore$ It is not transitive

15.(C) $\frac{{}^{20}C_2+{}^{19}C_2}{{}^{39}C_3}=\frac{19}{481}$

Number of APs = Number of ways of selecting two numbers whose sum is even.

16.(B) $\int\left[a^x \ln x+\underbrace{a^x \ln a}_{II}.\underbrace{x(\ln x-1)}_{I}\right]dx$

$=\int a^x.\ln x\,dx+\left[x(\ln x-1)a^x-\int\left[x.\frac{1}{x}+(\ln x-1)\right]a^x\right]dx$

$=\int a^x.\ln x\,dx+\left[x(\ln x/e)\right]a^x-\int(\ln x)a^x dx$

17.(A) $\left(2\sin^2 91^\circ-1\right)\left(2\sin^2 92^\circ-1\right)....\left(2\sin^2 180^\circ-1\right)$, In this product there exists a factor $\left(2\sin^2 135^\circ-1\right)$, which is equal to zero. Thus, the product of all terms is zero.

18.(B) Let $P\equiv\left(at_1^2,2at_1\right)$ and $Q\equiv\left(at_2^2,2at_2\right)$.

$\Rightarrow \quad t_1t_2=-1$

Equation of AP is $y=\frac{2}{t_1}x_1$.

$\Rightarrow R\equiv\left(-a,-\frac{2a}{t_1}\right)$

Similarly, $T\equiv\left(-a,\frac{-2a}{t_2}\right)$

Slope of RS = $\frac{\frac{(-2a)}{t_1}}{-2a}=\frac{1}{t_1}$

Slope of TS = $\frac{\frac{(-2a)}{t_2}}{-2a}=\frac{1}{t_2}$

Now, $\left(\frac{1}{t_1}\right)\times\left(\frac{1}{t_2}\right)=\frac{1}{t_1t_2}=-1$

$\Rightarrow \quad \angle RST=90^\circ$

19.(D) Clearly BC + AC should be min (for perimeter P to be minimum) Take image of A in y = 4, say it is $A'(2,7)$

$\therefore$ now AC = $A'C$

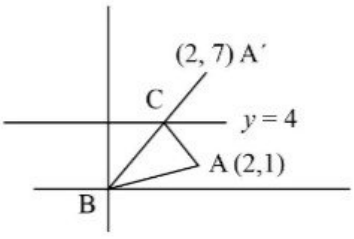
If $C=(t,4)$ then $\frac{4}{t}=\frac{7}{2}\Rightarrow t=\frac{8}{7}=\beta$ (Slope of BC = CA′)

for P to be max C should be at (4, 4)

i.e., $t=4$

$\Rightarrow \alpha=4$

$\therefore \quad =6\alpha+21\beta=24+24=48$



20.(D) $\bar{x}_{new} = \frac{4}{3}\bar{x}$

From given information $\bar{x}_{new} = \frac{n\bar{x} + 2k\bar{x}}{n+k} = \frac{4}{3}\bar{x}$

$\Rightarrow 3n + 6k = 4n + 4k$

$\Rightarrow 2k = n$

$\Rightarrow \frac{k}{n} = 0.5$

NUMERICAL TYPE

1.(997) Given, $f(x) = \frac{9^x}{9^x + 3}$ ... (i)

$\Rightarrow \quad f(1-x) = \frac{9^{1-x}}{9^{1-x} + 3}$

$\Rightarrow \quad f(1-x) = \frac{\frac{9}{9^x}}{\frac{9}{9^x} + 3} = \frac{9}{9 + 3 \cdot 9^x}$

$f(1-x) = \frac{9}{3(3 + 9^x)}$ ... (ii)

On adding Eqs. (i) and (ii), we get

$f(x) + f(1-x) = \frac{9^x}{9^x + 3} + \frac{9}{3(3 + 9^x)}$

$= \frac{3 \cdot 9^x + 9}{3(9^x + 3)} = \frac{3(9^x + 3)}{3(9^x + 3)}$

$\therefore \quad f(x) + f(1-x) = 1$ ... (iii)

Now, putting $x = \frac{1}{1996}, \frac{2}{1996}, \frac{3}{1996}, ..., \frac{998}{1996}$ in Eq. (iii),

we get $f\left(\frac{1}{1996}\right) + f\left(\frac{1995}{1996}\right) = 1$

$\Rightarrow \quad f\left(\frac{2}{1996}\right) + f\left(\frac{1994}{1996}\right) = 1$

$\Rightarrow \quad f\left(\frac{3}{1996}\right) + f\left(\frac{1993}{1996}\right) = 1$

.....

.....

$\Rightarrow \quad f\left(\frac{997}{1996}\right) + f\left(\frac{999}{1996}\right) = 1$

$$\Rightarrow \quad f\left(\frac{998}{1996}\right)+f\left(\frac{998}{1996}\right)=1$$

$$\Rightarrow \quad f\left(\frac{998}{1996}\right)=\frac{1}{2}$$

On adding all the above expressions, we get

$$k=f\left(\frac{1}{1996}\right)+f\left(\frac{2}{1996}\right)+\ldots+f\left(\frac{1995}{1996}\right)$$

$$=\underbrace{(1+1+1\ldots\ldots+1)}_{997 \text{ times}}+\frac{1}{2}$$

$$k=997+\frac{1}{2}$$

$$\therefore \quad \left|k-\frac{1}{2}\right|=\left|997+\frac{1}{2}-\frac{1}{2}\right|=997$$

2.(6) $2\left(x_2x_3+x_3x_1+x_1x_2\right)=\left(x_1+x_2+x_3\right)^2-\left(x_1^2+x_2^2+x_3^2\right)=-2$

$$x_1^3+x_2^3+x_3^3-3x_1x_2x_3$$

$$=\left(x_1+x_2+x_3\right)\left(x_1^2+x_2^2+x_3^2-x_2x_3-x_3x_1-x_1x_2\right)$$

$$=(2)(6+1)=14$$

$$\Rightarrow x_1x_2x_3=-2.$$

Equation whose roots are x_1, x_2, x_3 is $x^3-2x^2-x+2=0 \Rightarrow x=2,1,-1.$

Thus, $(x_2-x_3)(x_3-x_1)(x_1-x_2)=6.$

3.(7) Midpoint of the line segment is $\left(\frac{1-3}{2},\frac{2+6}{2},\frac{6+2}{2}\right)\equiv(-1,4,4)$

Parallel vector to the required line $=(-1+6)\hat{i}+(4-2)\hat{j}+(4-4)\hat{k}=5\hat{i}+2\hat{j}+0\hat{k}$

Hence, equation of the line is $\frac{x+6}{5}=\frac{y-2}{2}=\frac{z-4}{0}$

$$\Rightarrow l=5, m=2, n=0$$

4.(4) $(-\infty,-3]\cup(-2,-1)\cup(2,\infty)$

$$\log_{(|x|-1)}\left(x^2+4x+4\right)\geq 0$$

Case 1: $0<|x|-1<1$

i.e., $1<|x|<2$, then $0<x^2+4x+4\leq 1\Rightarrow x^2+4x+3\leq 0 \,\&\, (x+2)^2\neq 0$

$$\Rightarrow -3\leq x\leq -1 \,\&\, x\neq -2$$

So, $x\in(-2,-1)$

Case 2 : $|x|-1>1$

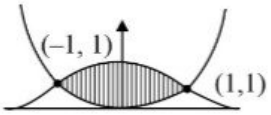
i.e., $|x| > 2$

$x^2 + 4x + 4 \geq 1 \Rightarrow (x+1)(x+3) \geq 0 \Rightarrow x \in (-\infty, -3] \cup [-1, \infty]$

$\Rightarrow x \in (-\infty, -3] \cup (2, \infty)$

Hence, domain is $(-\infty, -3] \cup (-2, -1) \cup (2, \infty)$

5.(3) Drawing graphs of $y = x^2$ *and* $y = \dfrac{2}{x^2+1}$



For points of intersection, $x^2 = \dfrac{2}{x^2+1} \Rightarrow x^4 + x^2 - 2 = 0$

$x^2 = 1 \Rightarrow x = \pm 1$

Required area $= 2\int_0^1 \left(\dfrac{2}{1+x^2} - x^2\right) dx = \left(\pi - \dfrac{2}{3}\right) sq.$ units

$K_1 = 1, K_2 = 2 \Rightarrow K_1 + K_2 = 3$

6.(9) $a-d, a, a+d, a-d+30$

If last three terms are in G.P. $(a+d)^2 = a(a-d+30)$

$\Rightarrow \quad a = \dfrac{d^2}{30-3d}$

$\Rightarrow$ For a to be integer $d = 9$

7.(4) The given equation can be written as $\dfrac{dy}{dx} - y\dfrac{\phi'(x)}{\phi(x)} = \dfrac{-y^2}{\phi(x)}$

$\Rightarrow \quad -\dfrac{1}{y^2}\dfrac{dy}{dx} + \dfrac{1}{y}\dfrac{\phi'(x)}{\phi(x)} = \dfrac{1}{\phi(x)}$

Put $\dfrac{1}{y} = u$ so that $-\dfrac{1}{y^2}\dfrac{dy}{dx} = \dfrac{du}{dx}$. Now (1) becomes $\dfrac{du}{dx} + u\dfrac{\phi'(x)}{\phi(x)} = \dfrac{1}{\phi(x)}$

This is a linear equation in u so its integrating factor is $e^{\int \frac{\phi'(x)}{\phi(x)} dx} = e^{\log \phi(x)} = \phi(x)$

Multiplying both sides by the integrating factor, we have

$\dfrac{d}{dx}[u\phi(x)] = 1 \Rightarrow u\phi(x) = x + \text{ constant} \Rightarrow y = \dfrac{\phi(x)}{x+C}$

$\Rightarrow \quad y(1) = \dfrac{1}{1+C} \Rightarrow C = 0.$ Thus $y = \dfrac{\phi(x)}{x}$

Hence $y(4) = \dfrac{1296}{4} = 324$

8.(100) Let $f(x)=\sin x+\cos x$. By considering the minimum and maximum value of

$$f(x) \text{ on } \left[0,\frac{\pi}{2}\right],\left[\frac{\pi}{2},\frac{3\pi}{4}\right],....\left[\frac{7\pi}{4},2\pi\right]$$

$$\left[f(x)\right]=\begin{cases}1 & 0\le x\le\dfrac{\pi}{2}\\ 0 & \dfrac{\pi}{2}<x\le\dfrac{3\pi}{4}\\ -1 & \dfrac{3\pi}{4}<x\le\pi\\ -2 & \pi<x<\dfrac{3\pi}{2}\\ -1 & \dfrac{3\pi}{2}\le x<\dfrac{7\pi}{4}\\ 0 & \dfrac{7\pi}{4}\le x<2\pi\end{cases}$$

So, $\int_0^{2\pi}[\sin x+\cos x]\,dx=\int_0^{\pi/2}dx+\int_{\pi/2}^{3\pi/4}0\,dx$

$$+\int_{3\pi/4}^{\pi}(-1)\,dx+\int_{\pi}^{3\pi/2}(-2)\,dx+\int_{3\pi/2}^{7\pi/4}(-1)\,dx+\int_{7\pi/4}^{2\pi}0\,dx$$

$$=\frac{\pi}{2}+\frac{3\pi}{4}-\pi+2\pi-3\pi+\frac{3\pi}{2}-\frac{7\pi}{4}=-\pi$$

Since $\sin x+\cos x$ is a periodic function with period 2π, $\int_0^{200\pi}[\sin x+\cos x]\,dx=-100\pi$

9.(4) For maximum number of common chords, we must have maximum number of common points. For y-coordinate of the point of intersections, we solve $4y+(y-7)^2=r^2$

or $\quad y^2-10y+49-r^2=0$

This equation will have two distinct real roots if $100-4\left(49-r^2\right)>0$

$\Rightarrow \quad r^2>24$

$\Rightarrow \quad r>\sqrt{24}$

$\Rightarrow$ least value of $[r]$ is 4.

10.(2) Since $g(x)=f^{-1}(x)$ so $f\left(g(x)\right)=(x)$ for all x. Differentiating $f'\left(g(x)\right)g'(x)=1$.

Let $g(1)=k$, then $f\left(g(1)\right)=f(k)=1$. Since $f(0)=1$ *and* f is continuous and increasing function so one-one thus $k=0$. Putting $x=1$ in (i), we get $f'(0)g'(1)=1$. But $f'(x)=3x^2+(1/2)e^{x/2}\Rightarrow f'(0)=\frac{1}{2}$

$$\therefore \quad \left(\frac{1}{2}\right)g'(1)=1\Rightarrow g'(1)=2.$$


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